

EDA Report – Automobile Dataset

Introduction

Objective of the Analysis

The objective of this exploratory data analysis (EDA) is to investigate the automobile dataset, identify patterns and relationship within the data, and gain insights into factors that influence automobile prices. The analysis includes data cleaning, preprocessing, visualisation and interpretation of key trends.

Dataset Description

The automobile dataset contains information about various vehicles, including manufacturer details, engine specifications, fuel type, horsepower, dimensions and price. These attributes provide valuable information for understanding the characteristics that contribute to vehicle pricing and market segmentation.

Data Cleaning

Prior to analysis, several data cleaning steps were performed to improve the quality and reliability of the dataset.

Removal of duplicate Rows

The dataset was examined for duplicate records using the `duplicated()` function. Any duplicate rows identified were removed to ensure that each observation represented a unique vehicle.

Removal of Missing Values

Missing values can affect statistical analysis and visualisations. Rows containing missing values were identified and removed using the `dropna()` function. This ensured that the analysis was conducted using complete and reliable data.

Data Type Conversion

Several columns containing numerical information were initially stored as text. These columns were converted to appropriate numeric data types using `pd.to_numeric()`. e.g. price, Horsepower, Peak RPM, Bore, Stroke, Normalized Losses. Converting these variables enabled accurate calculations, correlations and visualisations.

Which are the 5 most expensive cars?

How do the most expensive and cheapest cars compare? Exploring the most expensive cars highlights if some movie cars are worth the money spent on them based on their fuel economy (mpg or miles per gallon)

The five most expensive automobiles were identified and compared with the five least expensive vehicles based on fuel economy measurements. The analysis indicated that high prices cars generally offered lower fuel efficiency due to larger engines and performance-oriented designs. whereas cheaper cars tended to provide better mileage and economical operation.

Top Five expensive Cars:

The chart and table show the five most expensive cars in the dataset dominated by Mercedes-Benz and BMW models, with one Porsche.

Key findings:

Mercedes-Benz has the highest priced vehicles with prices of \$45,499 and \$40,960

BMW appears twice among the top five, priced at \$41,315 and \$36,880

Porsche is the fifth most expensive car, priced at \$37,028

The most expensive vehicle is a Mercedes-Benzes at \$ 55,400

Fuel Efficiency:

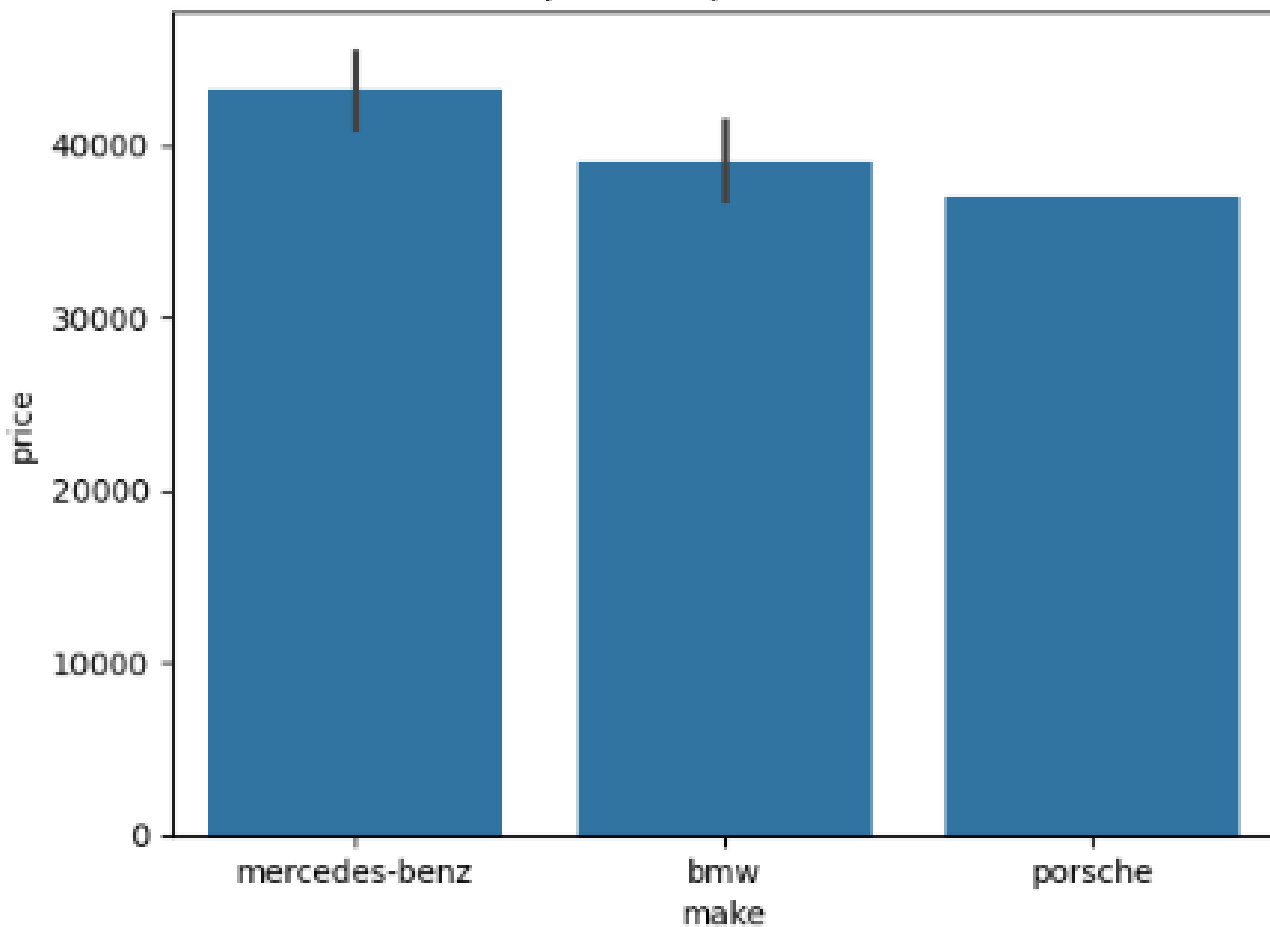
Fuel efficiency is relatively low which is typical for high performance luxury cars.

Mercedes-Benz models have the lowest fuel economy

Porsche provides the best fuel efficiency among the top prices cars

make	Price	Wheel-base	length	width	height	Curb-weight	Engine-size	Compression-ratio	City-mpg	Highway-mpg
Mercedes-Benzes	45400	112	199	72	55	3715	304	8	14	16
BMW	41315	103	193	67	53	3380	209	8	16	22
Mercedes-Benzes	40960	120	208	71	56	3900	308	8	14	16
Porsche	37028	89	168	65	51	2800	194	9	17	25
BMW	36880	110	197	70	56	3505	209	8	15	20

Top Five Expensive cars



Top Five Cheapest cars:

The chart highlights the five least expensive cars in the dataset, with prices ranging from approximately \$5,100 to \$5,400. These vehicles are primarily compact, lightweight, and fuel-efficient models designed for affordability and economy.

Key Findings:

Subaru is the cheapest vehicle at \$5,118

Chevrolet follows closely at \$5151

Mazda, Toyota and Mitsubishi complete the list wit prices between \$5195 and 5389

The price difference among the five cheapest cars is relatively small, indicating a highly competitive budget-car segment.

Fuel Efficiency:

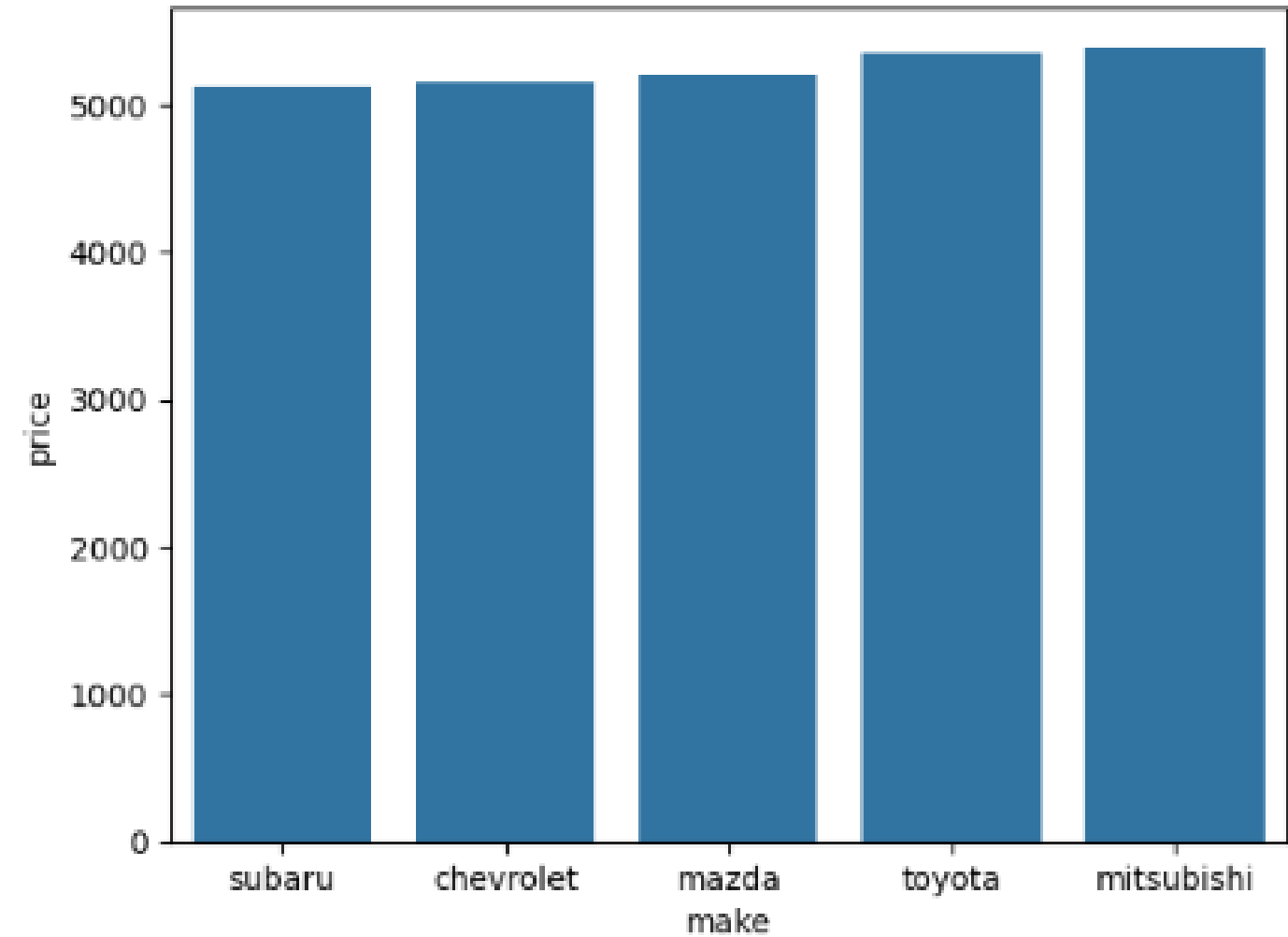
The cheapest cars offer excellent fuel economy.

Chevrettes achieves the highest efficiency

Toyota and Mitsubishi also perform well

Subaru and Mazda maintain strong fuel economy.

Top Five Cheapest cars



make	Price	Wheel-base	length	width	height	Curb-weight	Engine-size	Compression-ratio	City-mpg	Highway-mpg
Subaru	5118	93	156	63	53	2050	97	9	31	36
Chevrolet	5151	88	141	60	53	1488	61	9	47	53
Mazda	5195	93	159	64	54	1890	91	9	30	31
Toyota	5348	95	158	63	54	1985	92	9	35	39
Mitsubishi	5389	93	157	64	50	1918	92	9	37	41

Comparison between expensive and cheapest cars

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Price Distribution by Make Category

This chart compares the average prices of the most expensive and least expensive car manufacturers in the dataset.

Key Observations:

There is a substantial price gap between the expensive and cheapest car categories

Mercedes-Benz has the highest average price at approximately \$43,180 making it the most expensive brand in the analysis

BMW follows with an average price of approximately \$37,028

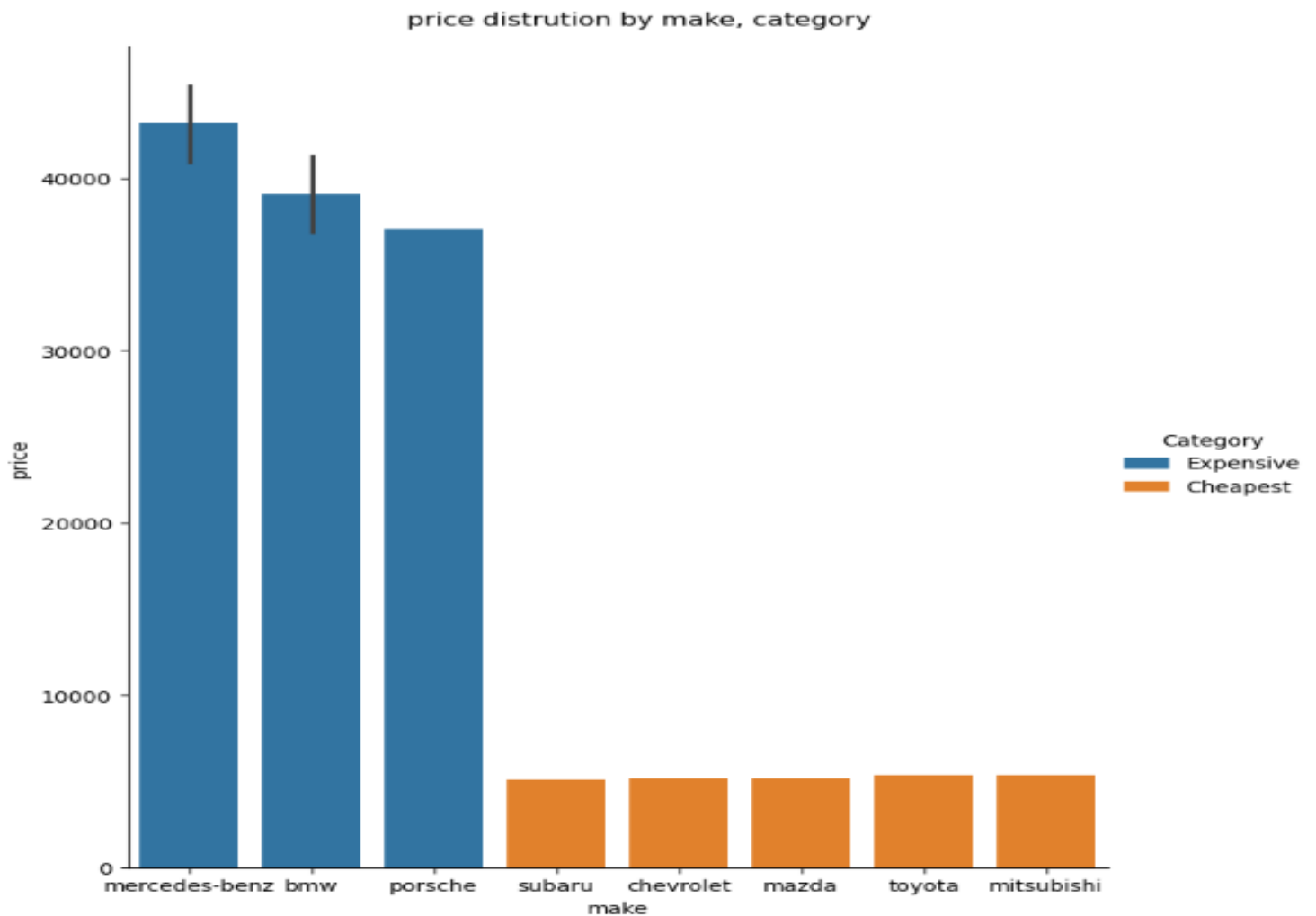
Among the cheapest brands Subaru has the lowest average price at \$5,118, while Mitsubishi has the highest among the budget brands at \$5,389

Insights:

Premium brands command significantly higher prices due to larger engines, greater vehicle dimensions, higher curb weights, and luxury features

Budget brands focus on affordability, compact design, fuel efficiency, and lower ownership costs.

The narrow price range among the cheapest bands suggested strong competition in the economy-car segment, whereas luxury brands show greater price variation.



Expensive vs Cheapest Cars

This chart compares the Highway fuel efficiency (MPG) of the most expensive and cheapest A clear inverse relationship between price and fuel economy is evident.

Key Findings

The most expensive cars have the lowest highway fuel efficiency.:

Mercedes-Benz: 16 MPG

BMW: 21 MPG (average)

Porche: 25 MPG

The cheapest cars achieve significantly higher highway MPG:

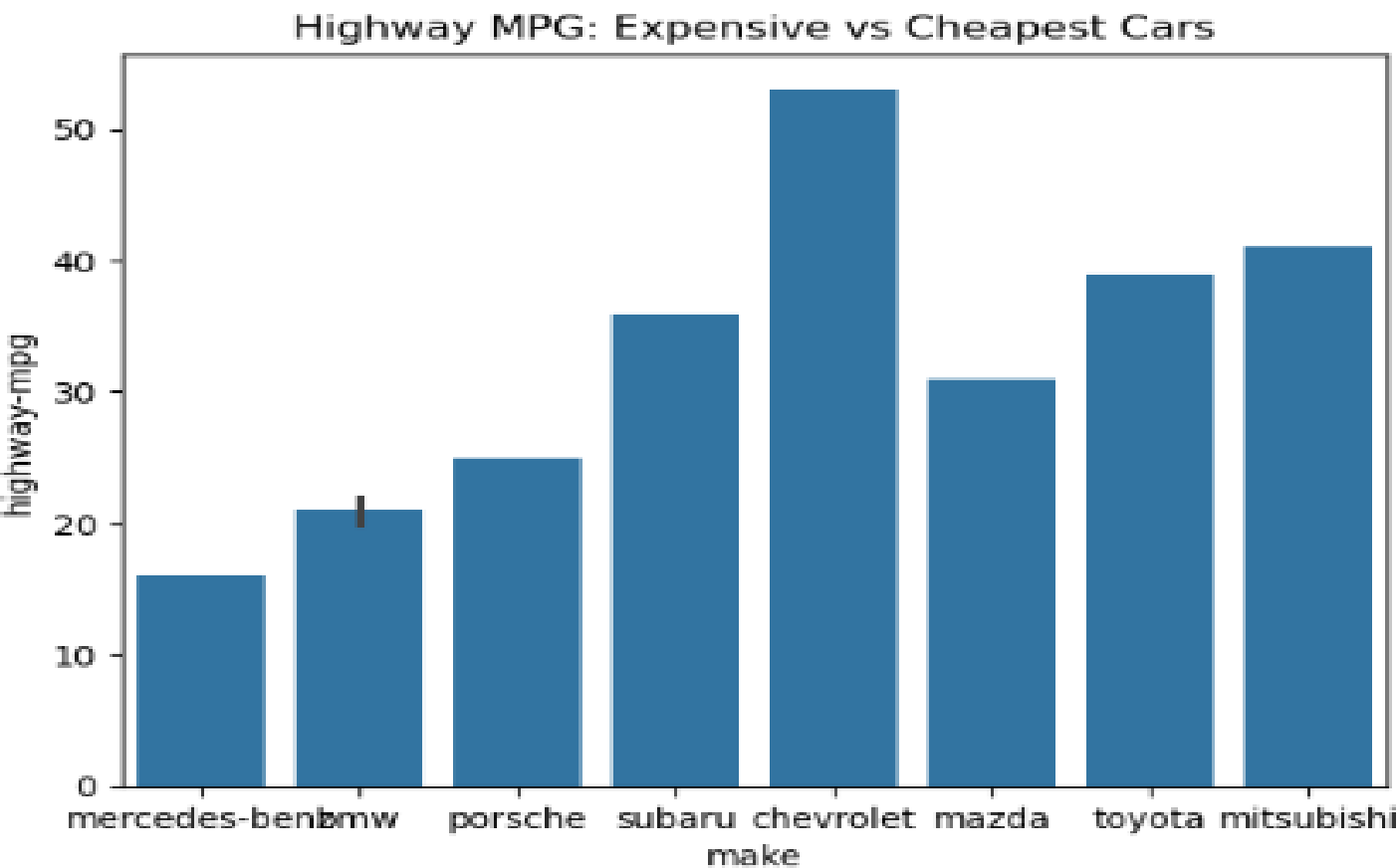
Chevrolet: 53 MPG (highest)

Mitsubishi: 41 MPG

Toyota: 39 MPG
Subaru: 36 MPG
Mazda: 31MPG

Comparison between Categories
Luxury vehicles average approximately 21 MPG on the highway
Budget vehicles average approximately 40 MPG on the highway

Insights
Expensive vehicle tends to have larger engines, heavier curb weights and higher performance capabilities, resulting in lower fuel efficiency.
Cheaper vehicles prioritize economy, lightweight design and smaller engines, leading to significantly better fuel consumption



Correlation Heatmap Analysis

The correlation heatmap reveals the relationships between vehicle characteristics and price highlights the key factors that influence car pricing and fuel efficiency.

Key Relationships with Price:

Curb Weight (0.95) has the strongest positive correlation with price, indicating that heavier vehicles tend to be more expensive

Engine Size (0.94) is also highly correlated with price, suggesting that larger engines significantly increase vehicle cost.

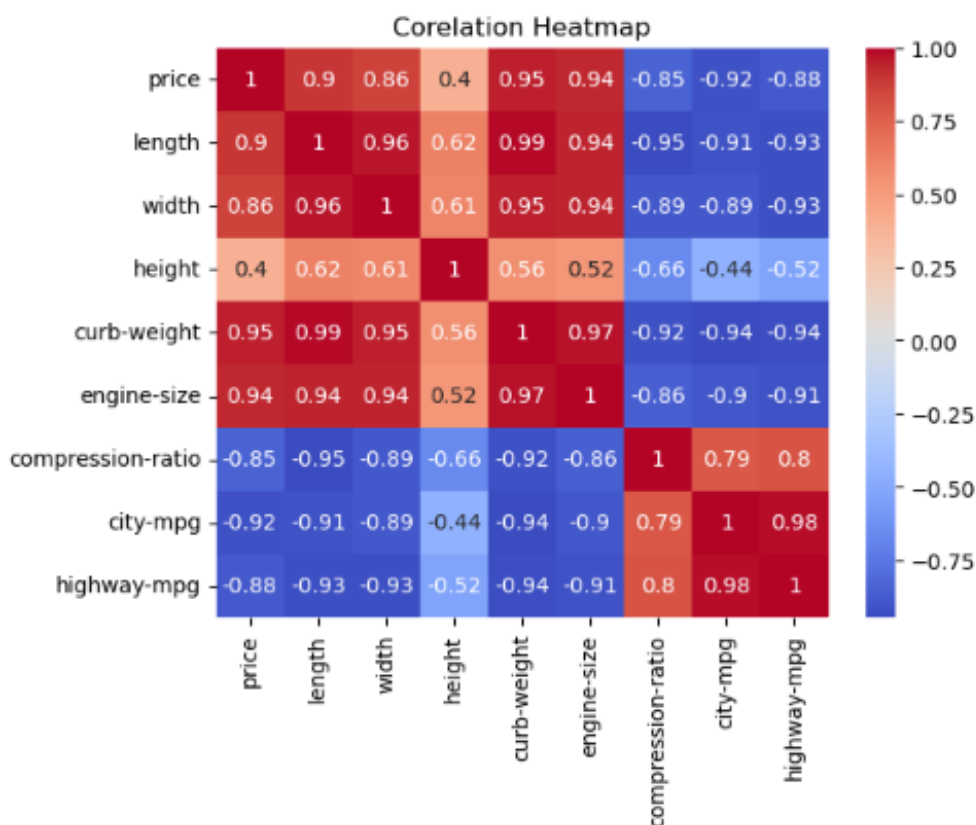
Length (0.90) and width (0.86) show strong positive correlations with price meaning larger vehicles generally command higher prices.

Height (0.40) has only a moderate positive correlation, indicating the vehicle height has less influence on price

Fuel Efficiency and price

City MPG (-0.92) and Highway MPG (-0.88) have strong negative correlation with price

This suggest that more expensive vehicles tend to be less fuel-efficient while cheaper vehicles generally achieve higher fuel economy



Which manufacturer builds the most fuel-efficient vehicles?

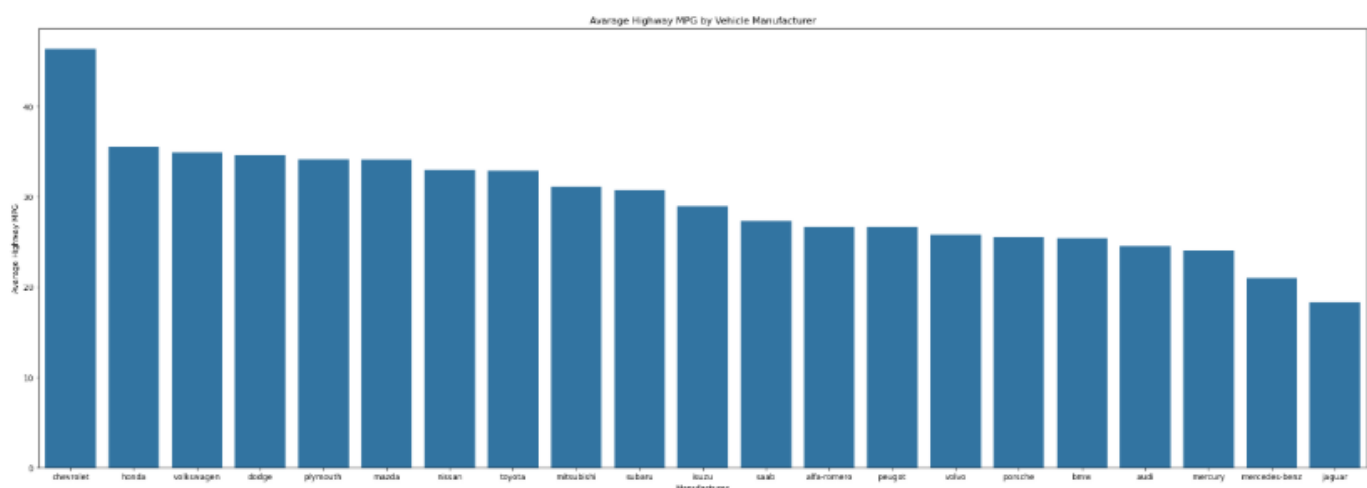
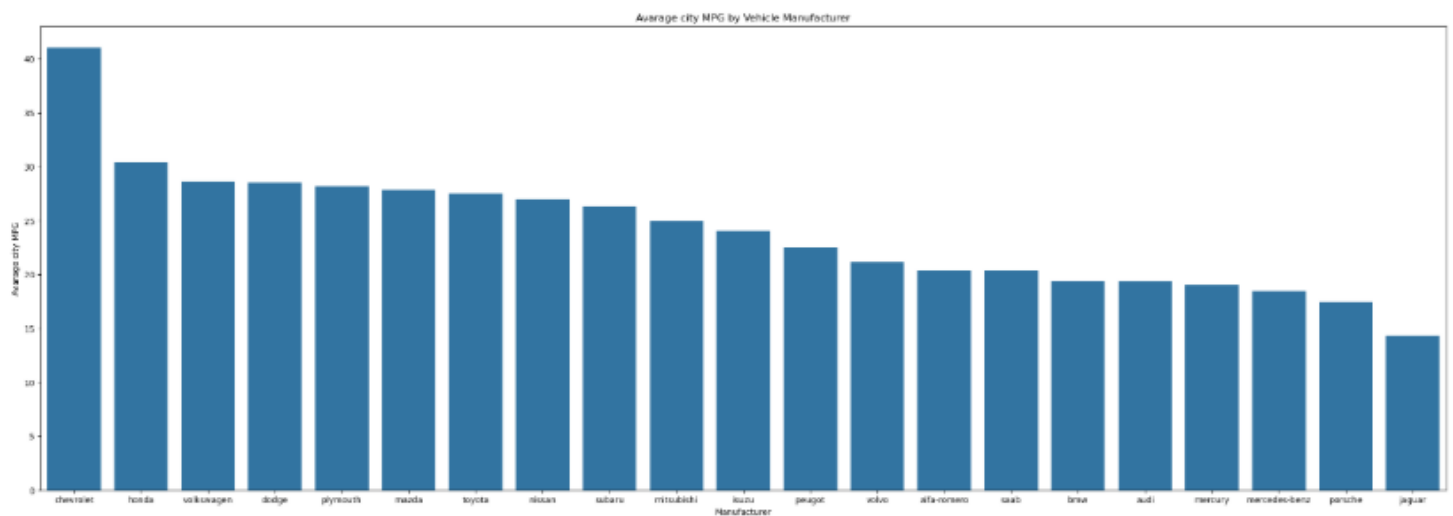
Compare the average mpg for each vehicle manufacturer's vehicles and create a bar plot

Manufacturers with the highest average MPG produce the most fuel-efficient vehicles.

Based on the dataset, Chevrolet is the most fuel-efficient manufacturer, followed by Mitsubishi Toyota and Subaru

Luxury manufacturers such as Mercedes-Benzes, BMW and Porsche have lower average MPG because their vehicles typically feature larger engines and heavier designs.

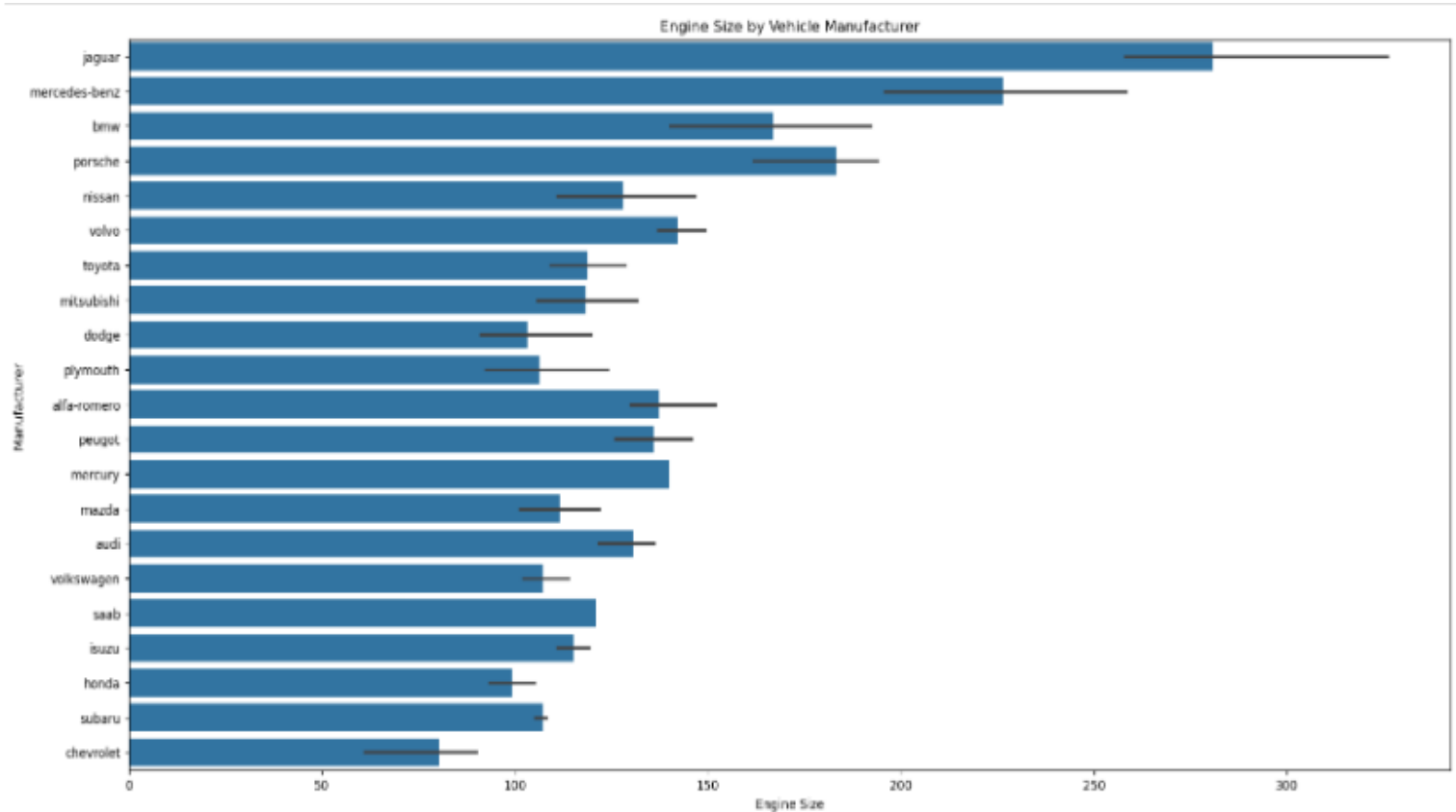
Overall economy car manufacturers deliver significantly better fuel efficiency than premium brands highlighting the trade off between performance and fuel economy



Which vehicles have the largest engine capacity.

Sort the dataframe based on the engine-size column.

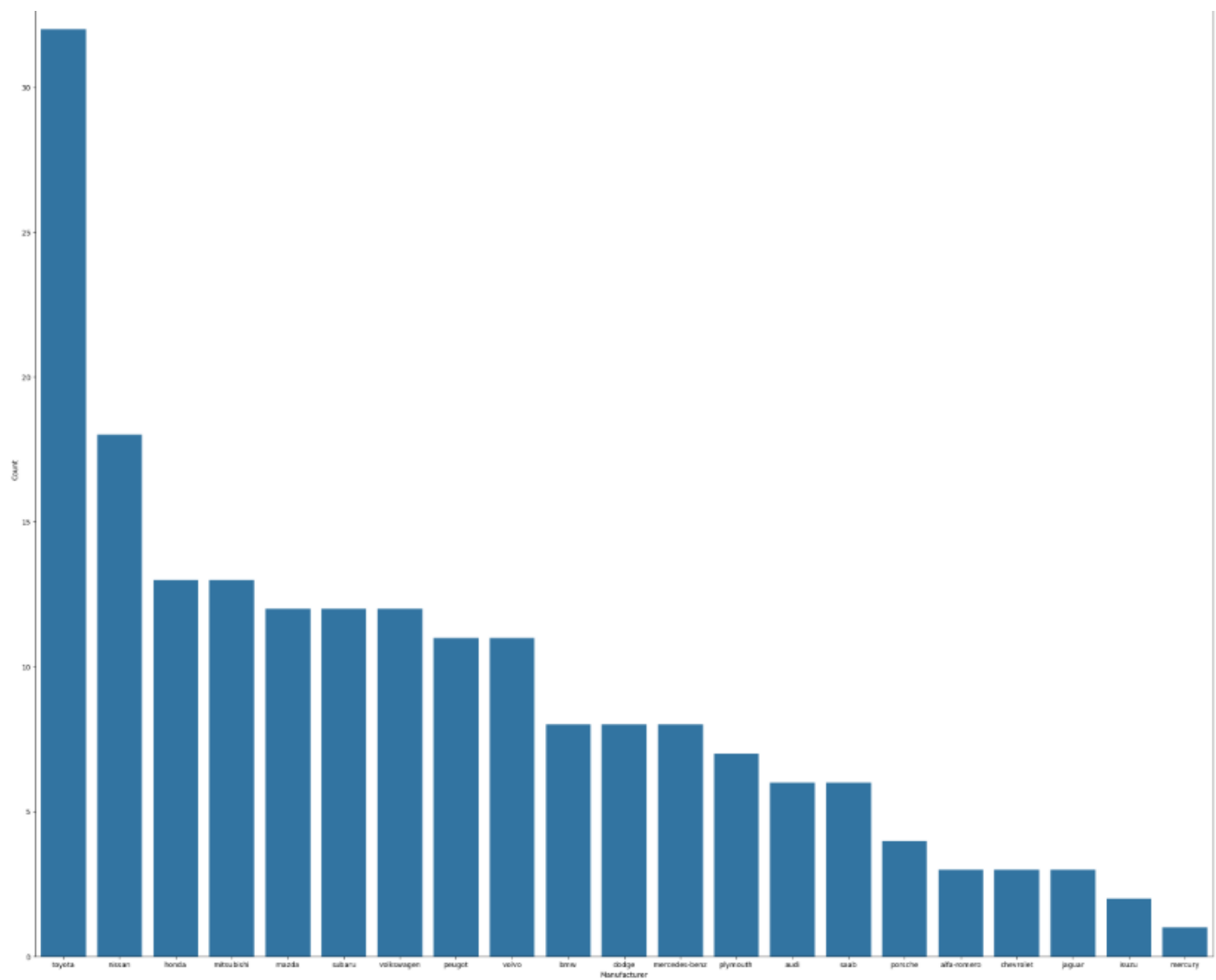
Vehicle with the largest engine capacities is typically Jaguar, Mercedes-Benz, followed by BMW and other luxury or performance brands.



Which vehicle manufacturer has the most car models in the dataset

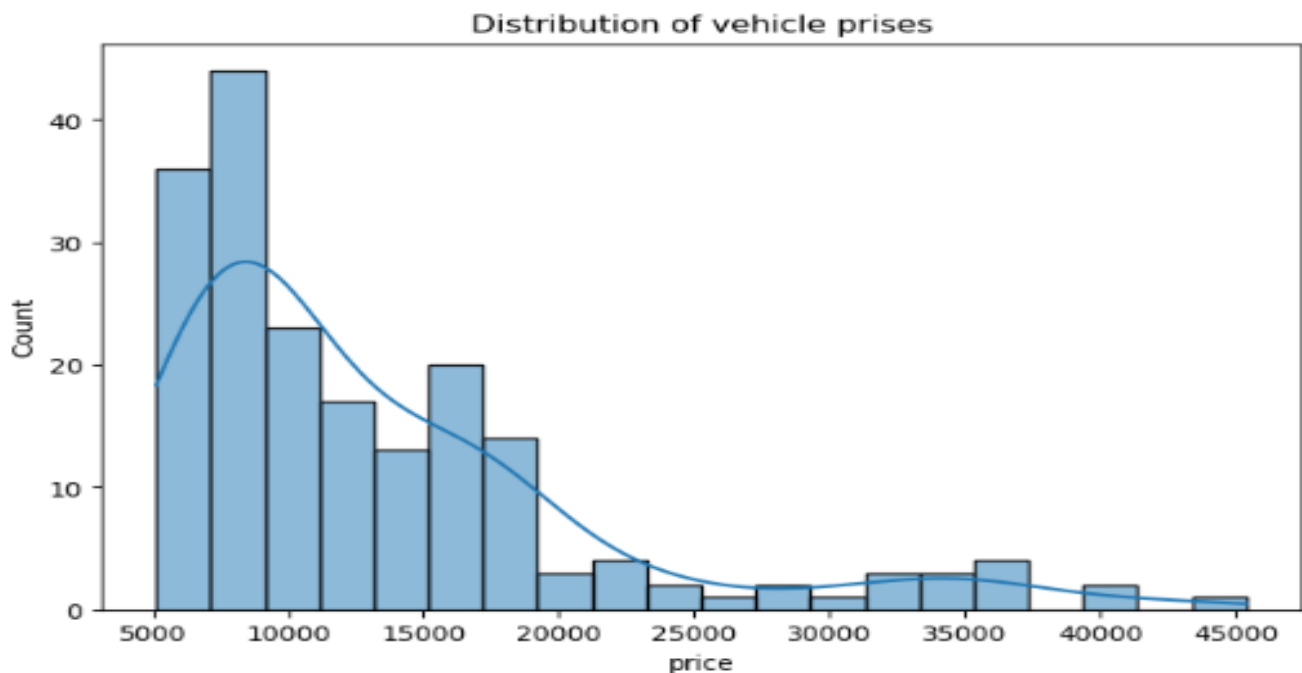
Toyota typically has the most car models followed by manufacturers such as Nissan, Honda, Mitsubishi, Mazda, Subaru

Manufacturer	Count
toyota	32
nissan	18
honda	13
mitsubishi	13
mazda	12
subaru	12
volkswagen	12
peugot	11
volvo	11
bmw	8
dodge	8
mercedes-benz	8
plymouth	7
audi	6
saab	6
porsche	4
alfa-romero	3
chevrolet	3
jaguar	3
isuzu	2
mercury	1



Exploratory Data Analysis

Price Distribution



The histogram shows that vehicle price is highly right-skewed (positively skewed). Most vehicles are concentrated in the lower price range (5000 to 15000), indicating that the dataset mainly consists of affordable or budget vehicles.

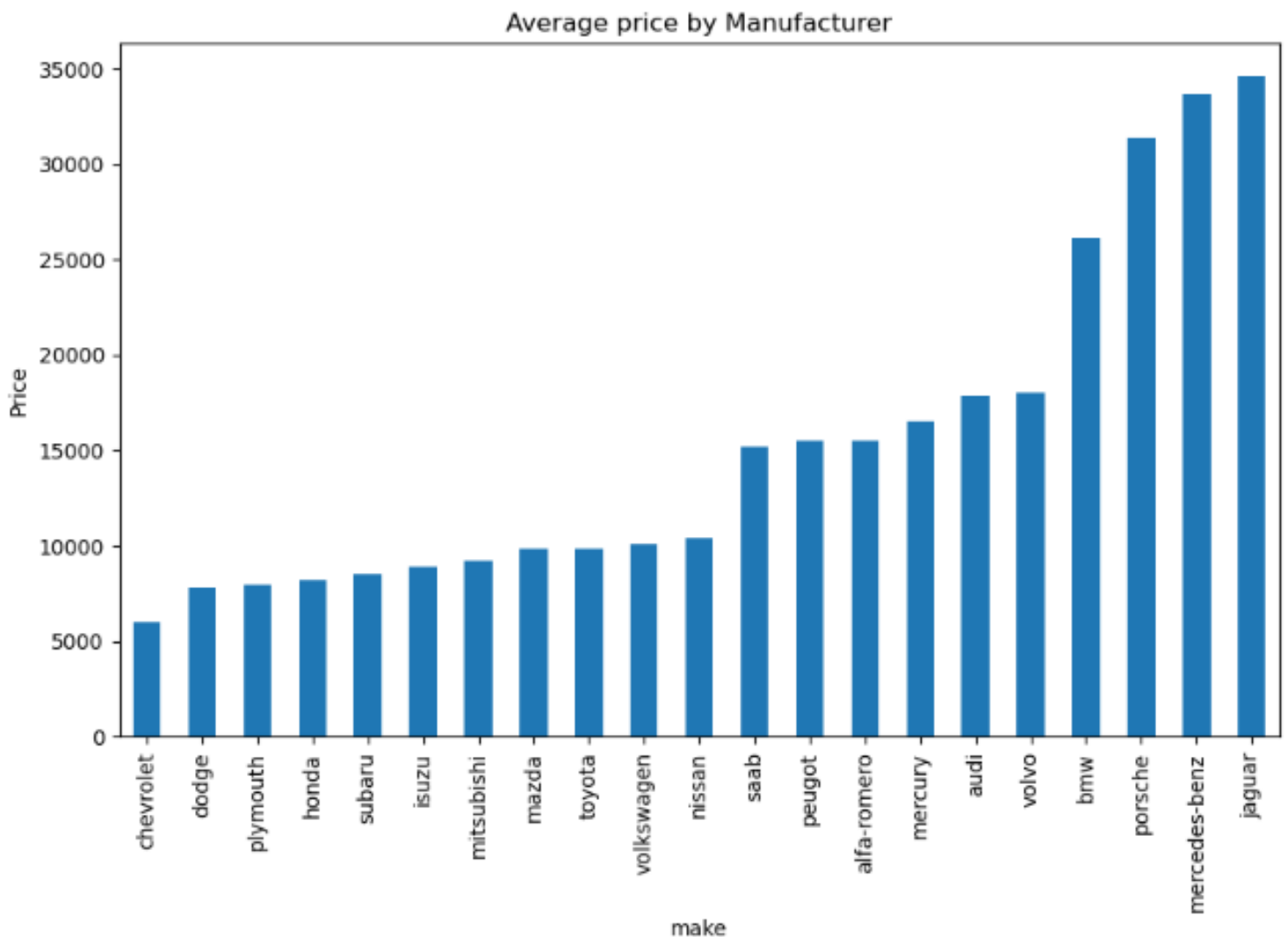
As the price increases the frequency of vehicles decreases sharply. There are relatively few mid-range vehicles (15000 to 25000) and only a small number of high-prices vehicles above 30000, with the distribution stretching up to around 45,000 forming a long right tail.

The KDE curves confirm this pattern showing a peak in the lower price range and a gradual decline as prices rise. This suggests the presence of outlier or luxury vehicles in the dataset,

Key Insights:

- 1) Majority of cars are low-priced (5k – 15k range)
- 2) Strong positive skew in price distribution
- 3) Presence of high-value outliers (30k – 45k range)
- 4) Data is not normally distributed, so transformations may be useful for modelling

Average price by Manufacturer



EDA Summary – Average Price by Manufacturer

The bar chart compares the average vehicle price across difficult manufacturers. It reveals significant variation in pricing, highlighting the distinction between economy and luxury vehicle brands.

Chevrolet has the lowest average vehicle price at approximately \$6000, followed by manufacturers such as Dodge, Plymouth, Honda, Subaru, Isuzu, Mitsubishi, Mazda, Toyota, Volkswagen and Nissan, whose average prices generally range between \$8000 and \$11000. These brands represent the more affordable segment of the market.

Manufacturers such as Saab, Peugeot, Alfa-Romeo, Mercury, Audi, and Volvo occupy the mid-range category, with average prices between \$15000 and \$18000.

The highest average prices are observed for luxury brands, with BMW, Porsche, Mercedes Benz, and Jaguar commanding significantly higher values. Jaguar has the highest average vehicle price at approximately \$35000 followed closely by Mercedes-Benz and Porsche.

Key Insights

- 1) Average vehicle prices vary substantially between manufacturers
- 2) Economy brands (Chevrolet, Dodge, Honda, Toyota, Nissan) have the lowest average prices.
- 3) Premium and luxury brands (BMW, Porsche, Mercedes-Benz, Jaguar) have significantly higher average prices.
- 4) Jaguar appears to be the most expensive manufacturer on average in this dataset
- 5) Manufacturer brand has a strong influence on vehicle pricing and could be an important predictor in price modelling.

The analysis indicates a clear relationship between manufacturer and vehicle price. Luxury manufacturers consistently command higher average prices than economy brands, suggesting that brand reputation, vehicle features and market positioning play a major role determining automobile prices. This variable is likely to be valuable for predictive modelling and market segmentation analyses

Horsepower vs Price

The scatter plot illustrates the relationship between horsepower and vehicle price. A clear positive correlation can be observed, indicating that vehicles with higher horsepower generally tend to have higher prices.

Most vehicles in the dataset are concentrated between 60 to 120 horsepower, with price ranging from \$5000 to 18000. As horsepower increases beyond this range, vehicle price tends to rise significantly, suggesting that engine performance is an important factor influencing automobile value

Key Insights

- 1) There is a strong positive relationship between horsepower and vehicle price.
- 2) Vehicles with lower horsepower (50-120 HP) are generally less expensive
- 3) Higher-horsepower vehicle tends to command higher market prices.
- 4) Several luxury vehicle outliers appear in the upper-right region of the plot with both high horsepower and high prices
- 5) Horsepower is likely to be an important predictor variable for price estimation models.

